

NUMERICAL METHODS

Interpolation with Central Differences

Stirling · Bessel · Everett · Gaussian Formulae

Objectives

To apply, and compare central-difference interpolation formulae to estimate function values and derivatives at points near the center of equally spaced data sets and to evaluate accuracy and numerical behavior.

Language

Up to you (Python / C++)

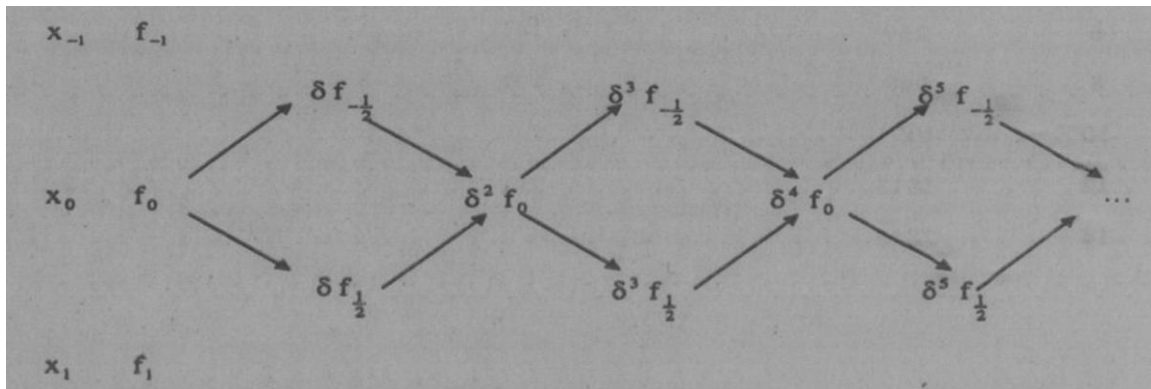
Series of formulae making use of central differences but with slightly different properties include:

- Stirling's Formula
- Bessel's Formula
- Everett's Formula
- Gaussian Formula
- Forward Formula
- Backward Formula

Stirling's Interpolation Formula

$$\begin{aligned}
 f_p = f_0 &+ \frac{1}{2}p \left(\delta f_{-\frac{1}{2}} + \delta f_{\frac{1}{2}} \right) + \frac{1}{2!}p^2 \delta^2 f_0 + \frac{p(p^2-1)}{2 \times 3!} \left(\delta^3 f_{-\frac{1}{2}} + \delta^3 f_{\frac{1}{2}} \right) \\
 &+ \frac{p^2(p^2-1)}{4!} \delta^4 f_0 + \frac{p(p^2-1)(p^2-4)}{2 \times 5!} \left(\delta^5 f_{-\frac{1}{2}} + \delta^5 f_{\frac{1}{2}} \right) \\
 &+ \frac{p^2(p^2-1)(p^2-4)}{6!} \delta^6 f_0 + \dots \quad \dots \quad (3.5(a))
 \end{aligned}$$

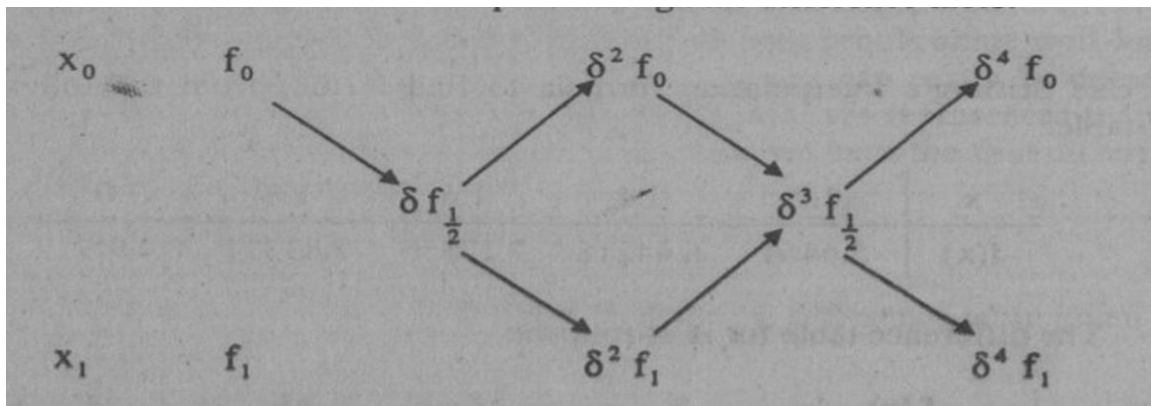
Path Followed by Stirling's Formula



Bessel's Interpolation Formula

$$f_p = f_0 + p\delta f_{\frac{1}{2}} + \frac{p(p-1)}{2 \cdot 2!} (\delta^2 f_0 + \delta^2 f_1) + \frac{p(p-1)(p-\frac{1}{2})}{3!} \delta^3 f_{\frac{1}{2}} + \frac{(p+1)p(p-1)(p-2)}{2 \cdot 4!} (\delta^4 f_0 + \delta^4 f_1) + \dots$$

Path Followed by Bessel's Formula



Everett's Interpolation Formula

$$f_p = q f_0 + \frac{q(q^2-1)}{3!} \delta^2 f_0 + \frac{q(q^2-1)(q^2-4)}{5!} \delta^4 f_0 + \dots + p f_1 + \frac{p(p^2-1)}{3!} \delta^2 f_1 + \frac{p(p^2-1)(p^2-4)}{5!} \delta^4 f_1 + \dots$$

where $q = 1 - p$

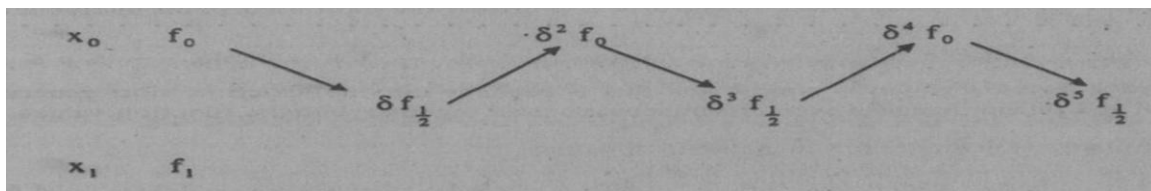
Path Followed by Everett's Formula

x_0	f_0	_____	$\delta^2 f_0$	_____	$\delta^4 f_0$	...
x_1	f_1	_____	$\delta^2 f_1$	_____	$\delta^4 f_1$	...

Gaussian Forward Interpolation Formula

$$f_p = f_0 + p\delta f_{\frac{1}{2}} + \frac{p(p-1)}{2!}\delta^2 f_0 + \frac{p(p+1)(p-1)}{3!}\delta^3 f_{\frac{1}{2}} + \frac{(p+1)p(p-1)(p-2)}{4!}\delta^4 f_0 + \dots$$

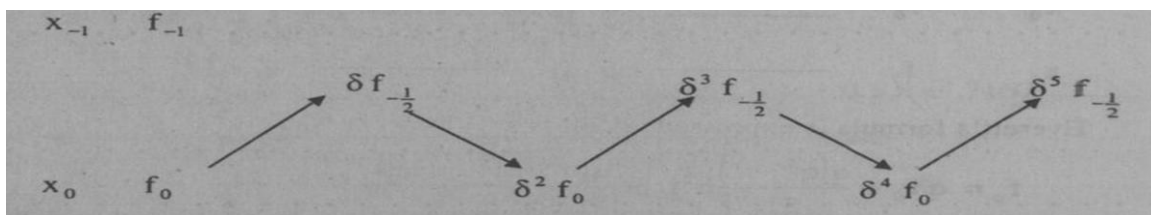
Path Followed by Gaussian Forward Interpolation Formula



Gaussian Backward Interpolation Formula

$$f_p = f_0 + p\delta f_{-\frac{1}{2}} + \frac{p(p+1)}{2!}\delta^2 f_0 + \frac{p(p+1)(p-1)}{3!}\delta^3 f_{-\frac{1}{2}} + \frac{(p+1)p(p-1)(p-2)}{4!}\delta^4 f_0 + \dots$$

Path Followed by Gaussian Backward Interpolation Formula



Things to Remember

- Mean of Gaussian forward and backward difference formula results in Stirling's formula.
- Stirling's formula is suitable for small values of p ($-0.25 < p < 0.25$).
- Bessel's formula is most famous of all. Suitable for values of p not far from 0.5. Can follow the range of Stirling's formula.
- Everett's formula is simple and fast and make use of even differences.

Task

Write a program that:

- Accepts n equally spaced data points along with functional values from the user.
- Builds the difference table and prints it in a tabular format.
- Compute the results of interpolation by making use of central differences applying different central difference interpolation formulae.
- Compare the results obtained by different formulae.